PRAKTIKUM BASIS DATA

MODUL 8

SCRIPTING SQL (CREATE INDEX, CREATE VIEW, ANALYZE TABLE)

Disusun Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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1. **MEMBUAT DESAIN DATABASE UNTUK SUATU PERUSAHAAN :**

a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.

QUERY

Worksheet | Query Builder

```
CREATE TABLE department (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(100),
    manager_id INT,
    location_id INT
);
```

OUTPUT

```
Table DEPARTMENT created.
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	DEPARTMENT_ID	NUMBER(38,0)	No	(null)	1	(null)
2	DEPARTMENT_NAME	VARCHAR2(100 BYTE)	Yes	(null)	2	(null)
3	MANAGER_ID	NUMBER(38,0)	Yes	(null)	3	(null)
4	LOCATION_ID	NUMBER(38,0)	Yes	(null)	4	(null)

b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL

QUERY

```
CREATE TABLE employee (
    employee_id INT PRIMARY KEY,
    last_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    salary DECIMAL(10,2),
    commission_pct DECIMAL(5,2),
    hire_date DATE NOT NULL,
    department_id INT,
    FOREIGN KEY (department_id) REFERENCES department(department_id)
);
```

OUTPUT

```
Table EMPLOYEE created.
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	EMPLOYEE_ID	NUMBER(38,0)	No	(null)	1	(null)
2	LAST_NAME	VARCHAR2(100 BYTE)	No	(null)	2	(null)
3	EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)
4	SALARY	NUMBER(10,2)	Yes	(null)	4	(null)
5	COMMISSION_PCT	NUMBER(5,2)	Yes	(null)	5	(null)
6	HIRE_DATE	DATE	No	(null)	6	(null)
7	DEPARTMENT_ID	NUMBER(38,0)	Yes	(null)	7	(null)

c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

QUERY

```
CREATE TABLE supplier (
    supplier_id INT PRIMARY KEY,
    supplier_name VARCHAR(100),
    contact_email VARCHAR(100)
);
```

OUTPUT

```
Table SUPPLIER created.
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	SUPPLIER_ID	NUMBER(38,0)	No	(null)	1	(null)
2	SUPPLIER_NAME	VARCHAR2(100 BYTE)	Yes	(null)	2	(null)
3	CONTACT_EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	3	(null)

QUERY

```
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    price DECIMAL(10,2),
    supplier_id INT,
    FOREIGN KEY (supplier_id) REFERENCES supplier(supplier_id)
);
```

OUTPUT

```
Table PRODUCT created.
```

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMME
1	PRODUCT_ID	NUMBER(38,0)	No	(null)	1	(null)
2	PRODUCT_NAME	VARCHAR2(100 BYTE)	Yes	(null)	2	(null)
3	PRICE	NUMBER(10,2)	Yes	(null)	3	(null)
4	SUPPLIER_ID	NUMBER(38,0)	Yes	(null)	4	(null)

2. ANALISIS TABLE-TABLE DENGAN COMPUTE STATISTICS

QUERY

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
```

OUTPUT

```
Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.
```

3. ISIKAN DATA-DATA YANG BERSESUAIAN DENGAN TABEL-TABEL PADA NOMOR 1 (MASING-MASING 10 RECORD).

a. Department

QUERY

```
INSERT INTO department VALUES (10, 'HR', 101, 1);
INSERT INTO department VALUES (20, 'IT', 102, 2);
INSERT INTO department VALUES (30, 'Finance', 103, 3);
INSERT INTO department VALUES (40, 'Marketing', 104, 4);
INSERT INTO department VALUES (50, 'Sales', 105, 5);
INSERT INTO department VALUES (60, 'Support', 106, 6);
INSERT INTO department VALUES (70, 'Logistics', 107, 7);
INSERT INTO department VALUES (80, 'RnD', 108, 8);
INSERT INTO department VALUES (90, 'Admin', 109, 9);
INSERT INTO department VALUES (100, 'Legal', 110, 10);
```

OUTPUT

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	HR	101	1
2	20	IT	102	2
3	30	Finance	103	3
4	40	Marketing	104	4
5	50	Sales	105	5
6	60	Support	106	6
7	70	Logistics	107	7
8	80	RnD	108	8
9	90	Admin	109	9
10	100	Legal	110	10

b. Employee

QUERY

```
INSERT INTO employee VALUES (1, 'Restu1', 'restu1@mail.com', 5000, 0.1, TO_DATE('2022-01-01','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (2, 'Restu2', 'restu2@mail.com', 6000, 0.2, TO_DATE('2022-02-01','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (3, 'Restu3', 'restu3@mail.com', 5500, 0.15, TO_DATE('2022-03-01','YYYY-MM-DD'), 20);
INSERT INTO employee VALUES (4, 'Restu4', 'restu4@mail.com', 7000, 0.1, TO_DATE('2022-04-01','YYYY-MM-DD'), 30);
INSERT INTO employee VALUES (5, 'Restu5', 'restu5@mail.com', 4500, 0.05, TO_DATE('2022-05-01','YYYY-MM-DD'), 40);
INSERT INTO employee VALUES (6, 'Restu6', 'restu6@mail.com', 8000, 0.2, TO_DATE('2022-06-01','YYYY-MM-DD'), 50);
INSERT INTO employee VALUES (7, 'Restu7', 'restu7@mail.com', 9000, 0.25, TO_DATE('2022-07-01','YYYY-MM-DD'), 60);
INSERT INTO employee VALUES (8, 'Restu8', 'restu8@mail.com', 4000, 0.05, TO_DATE('2022-08-01','YYYY-MM-DD'), 70);
INSERT INTO employee VALUES (9, 'Restu9', 'restu9@mail.com', 7500, 0.2, TO_DATE('2022-09-01','YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (10, 'Restu10', 'restu10@mail.com', 6500, 0.1, TO_DATE('2022-10-01','YYYY-MM-DD'), 90);
```

OUTPUT

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	1	Restu1	restu1@mail.com	5000	0,1	01-01-2022	80
2	2	Restu2	restu2@mail.com	6000	0,2	01-02-2022	80
3	3	Restu3	restu3@mail.com	5500	0,15	01-03-2022	20
4	4	Restu4	restu4@mail.com	7000	0,1	01-04-2022	30
5	5	Restu5	restu5@mail.com	4500	0,05	01-05-2022	40
6	6	Restu6	restu6@mail.com	8000	0,2	01-06-2022	50
7	7	Restu7	restu7@mail.com	9000	0,25	01-07-2022	60
8	8	Restu8	restu8@mail.com	4000	0,05	01-08-2022	70
9	9	Restu9	restu9@mail.com	7500	0,2	01-09-2022	80
10	10	Restu10	restu10@mail.com	6500	0,1	01-10-2022	90

4. ANALISIS TABLE-TABLE DENGAN COMPUTE STATISTICS

QUERY

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
```

OUTPUT

```
Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.
```

5. **BUAT VIEW EMPVU80 YANG BERISI ID_NUMBER, NAME, SALARY, DEPARTMENT_ID DARI PEGAWAI YANG BEKERJA DI DEPARTMENT = 80.**

QUERY

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

OUTPUT

```
View EMPVU80 created.
```

6. **KEMUDIAN TAMPILKAN STRUKTUR DARI VIEW EMPVU80.**

QUERY

```
DESCRIBE empvu80;
```

OUTPUT

```
Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER(38)
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER(38)
```

7. **BUAT NON-UNIQUE INDEX PADA KOLOM FOREIGN KEY YANG ADA PADA TABEL EMPLOYEE.**

QUERY

```
CREATE INDEX idx_dept_id
ON employee(department_id);
```

OUTPUT

```
Index IDX_DEPT_ID created.
```